

3. THE CELL

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- The cell is the basic unit of structure and functions in organism.
- The cells make up the structures of the living organisms and are responsible for carrying out the various biological processes.
- **Unicellular** organisms comprise of only one cell, *e.g. amoeba, Paramecium, and plasmodium.*
- **Multicellular** organisms are made up of many cells, *e.g. moulds, grasshopper, moss, eucalyptus, man and elephant.*
- A cell being a very small structure that cannot be seen with the naked eye, hence it requires the use of a powerful magnifying instrument known as the *microscope.*

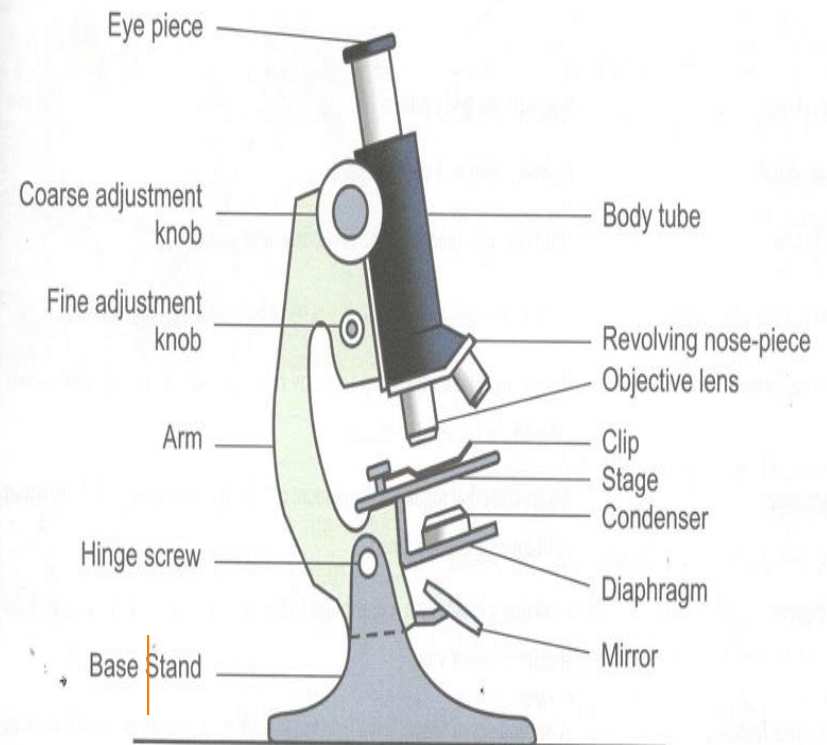
Functions of a microscope.

1. **Magnification-** making very small organisms to appear bigger so that they can be seen.
2. **Resolution-** ability to distinguish two structures that are very close together as distinct entities.

Types of microscope.

1. **Light microscope-** It uses light for illumination of the specimen to be viewed.
2. **Electron microscope-** The electron microscope uses an **electron beam** instead of light for illumination of the specimen to be viewed.

LIGHT MICROSCOPE



Uses of different parts of a light microscope.

1. **Arm/ limb-** it supports the body tube and stage.
2. **Base/ stand-** it provides a firm and stable support.
3. **Body tube-** it holds the eye-piece and revolving nose piece.
4. **Eye piece-** it contains a lens which contributes to the magnification of the image of the specimen under view.

5. **Coarse adjustment knob-** it brings image into rough focus (by raising and lowering the body tube through long distances).
6. **Fine adjustment knob-** it brings the image into sharp focus (by raising and lowering the body tube through smaller distances).
7. **Diaphragm-** it is an aperture that regulates the amount of light passing through the condenser to illuminate the specimen.

8. **Objective lenses-** they contribute to the magnification of the image of the specimen.
9. **Mirror-** it reflects light through the condenser onto the stage.
10. **Revolving nose piece-** it holds the objective lens in place thus enabling change from one objective lens to another.
11. **Condenser-** it concentrates light onto the stage.
12. **Stage-** it is a flat platform where specimen on the slide is placed.
13. **Clip-** it holds the slide in position.

HANDLING AND CARE OF A LIGHT MICROSCOPE.

1. Always use both hands when carrying a microscope.
- One hand should hold the base to provide support while the other hand holds the limb/arm.
2. Never place the microscope too close to the edge of the working bench or table as it can easily fall off.
3. Do not touch the mirror and the lenses with your fingers as it can make them wet or dirty.
4. Dirty lenses should be cleaned using a special soft lens tissue paper or tissue paper moistened with ethanol to avoid scratches.
5. Other parts of the microscope may be cleaned using a soft cloth or tissue paper.
6. Do not wet any part of the microscope as it can cause rusting.
7. Make sure the low power objective lens clicks into position in line with the eye-piece before and after use.
8. After use, always clean and store the microscope in a safe place, free from moisture and dust.

USING THE LIGHT MICROSCOPE

1. Place the microscope on the bench with the stage facing away from you.
2. Turn the low power objective lens until it clicks into position.
3. Ensure that the diaphragm is fully open.
4. Look through the eye-piece with one eye. Meanwhile adjust the mirror under the stage to ensure that maximum light can pass through. The circular area seen is referred to as the **field of view**.
5. Place the slide containing the specimen on the stage and clip it into position. Make sure that it is in the centre of the field of view.
6. Again, look through the eye-piece while adjusting the mirror under the stage to ensure that sufficient light is passing through the specimen.
7. Use the coarse adjustment knob to bring the low power objective lens to the lowest point. Viewing through the eye-piece, turn the coarse adjustment knob gently until the specimen comes into focus.

8. Use the fine adjustment knob to bring the image into sharp focus.
 9. Make a drawing of what you see.
 10. For higher magnifications, turn the medium power objective lens into position and adjust the focus using the coarse adjustment knob. For sharper images, use the fine adjustment knob.
 11. If finer details are required, turn the high power objective lens into position. Now use only the fine adjustment knob to bring details into sharper focus.
- Sometimes a camera can be fixed at the eye piece lens in order to take a photograph of the specimen. The photograph is called *photomicrograph*.

Study questions

1. **How is the low objective lens manipulated to focus a specimen for observation under a light microscope.?**

- ☐ Click the low power objective lens in position (bring it down to the lowest level using coarse adjustment knob)
- ☐ With the eyes on the two eye piece lens and using the coarse adjustment knob gradually raise/the low power objective lens to bring the specimen into focus.

2. Give a reason why only the fine adjustment knob should be used when using high power objective/ coarse adjustment knob should not be used to lower the high power objective lens.

- i. **To avoid breaking the slide, cover slip and lens.**
- ii. **To avoid destroying the specimen/ dirtfying the lens.**

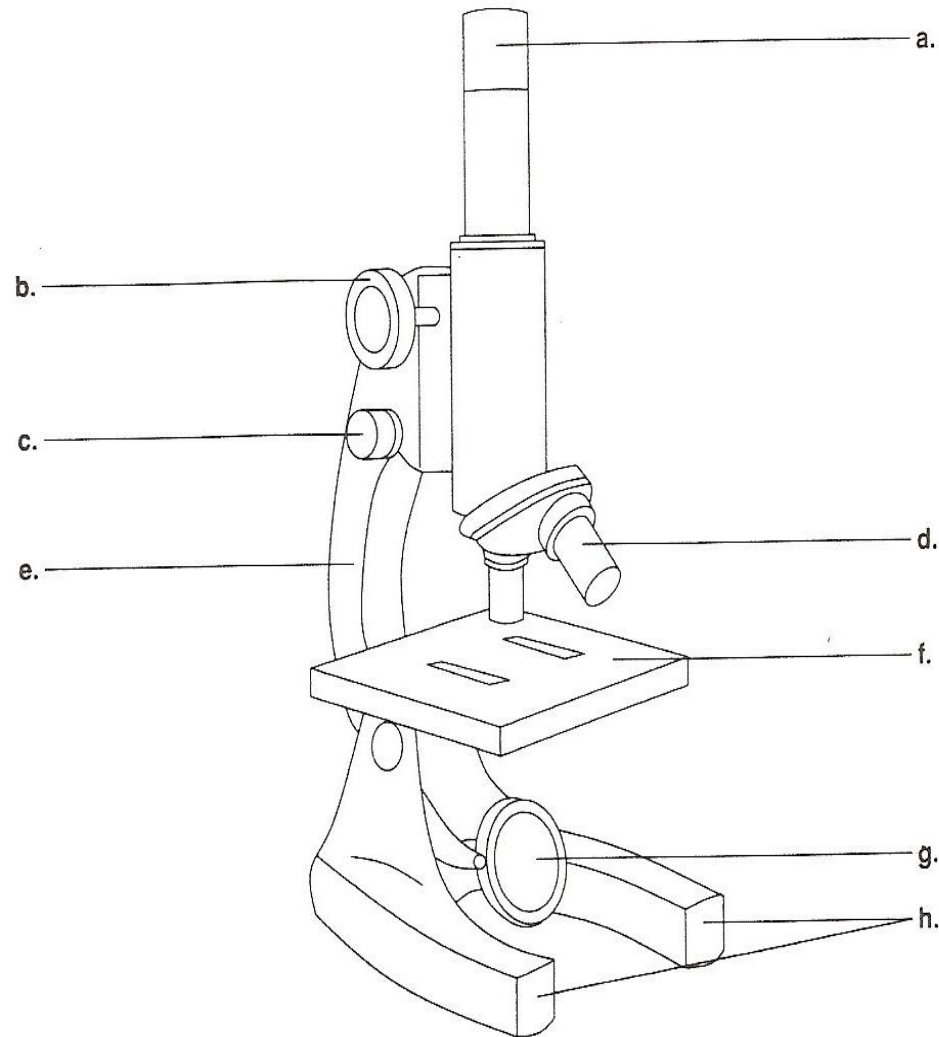
Differences between the light and electron microscopes

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Light microscope	Electron microscope
1. Uses light to illuminate the specimen.	1. Uses a beam of electrons to illuminate the specimen.
2. Uses glass lenses for magnification.	2. Uses magnetic lenses for magnification.
3. Has low resolving power.	3. Has very high resolving power.
4. Has low magnification power.	4. Has very high magnification power.
5. The specimen under view can be living or dead.	5. The specimen under view is dead.
6. The specimen is stained using normal dyes.	6. The specimen is stained using complex stains.
7. The specimen is mounted on a slide and placed on the stage in the open.	7. The specimen is mounted on a grid and placed in a vacuum.

Study question. Name the parts of the light microscope shown below.

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- a. Eyepiece
- b. Coarse Adjustment
- c. Fine Adjustment
- d. Objectives (LP, HP)
- e. Arm
- f. Stage
- g. Light source
- h. Base
- i. Diaphragm

Magnification of a light microscope.

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- The magnification of the image of the object viewed is calculated by multiplying the eye piece lens magnification by the objective lens magnification i.e.

Magnification = eye piece lens
magnification x objective lens
magnification.

Example

If the eye-piece lens has a magnification of X5 and the low power objective lens has a magnification of X10, then the total magnification is $5 \times 10 = X50$.

Study question.

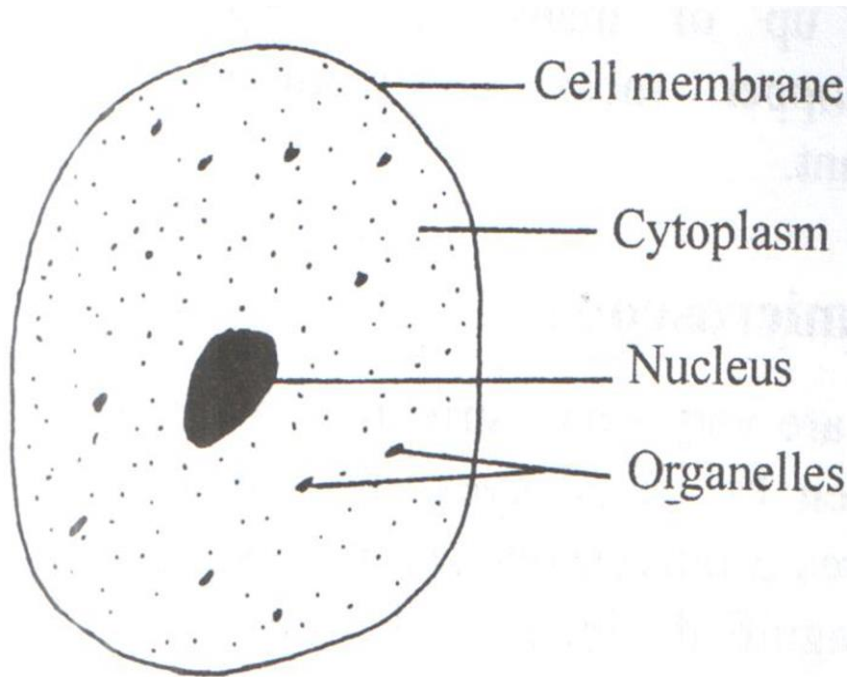
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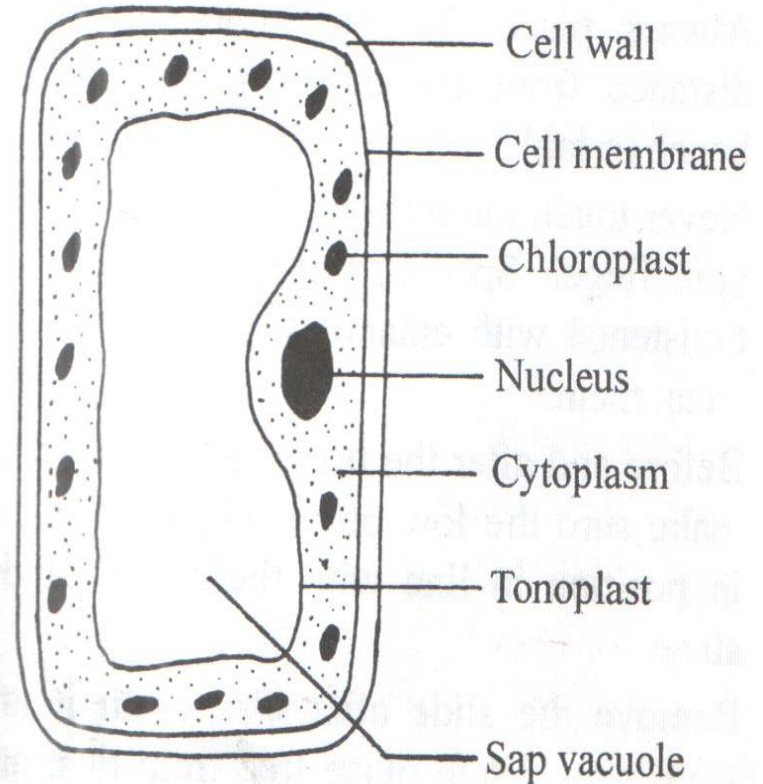
Eye-piece lens magnification.	Objective lens magnification.	Total magnification.
X5	X4	X20
X10	X5	
X10		X100
	X40	X600
X10	X100	

PLANT AND ANIMAL CELLS AS SEEN UNDER THE LIGHT MICROSCOPE.

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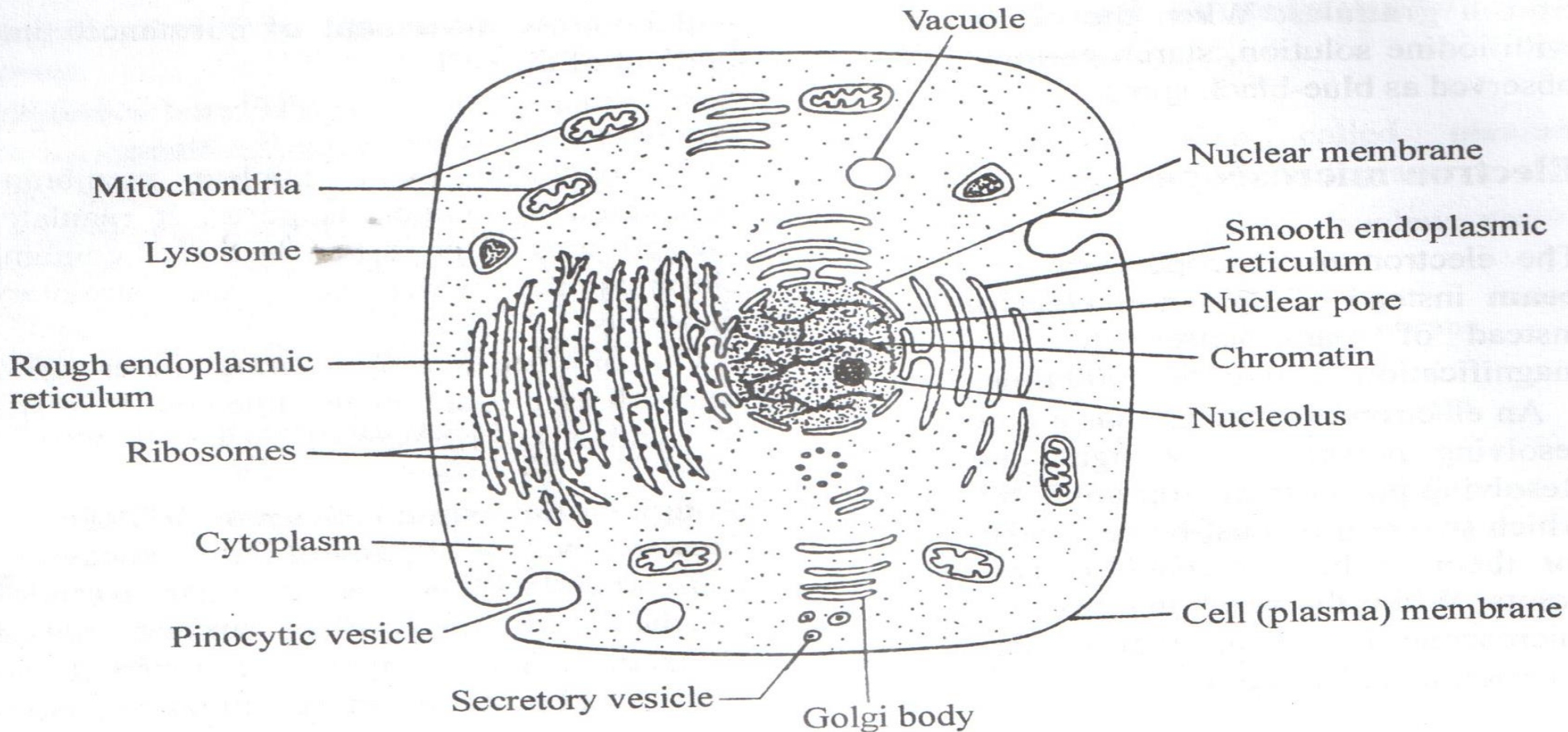
An animal cell



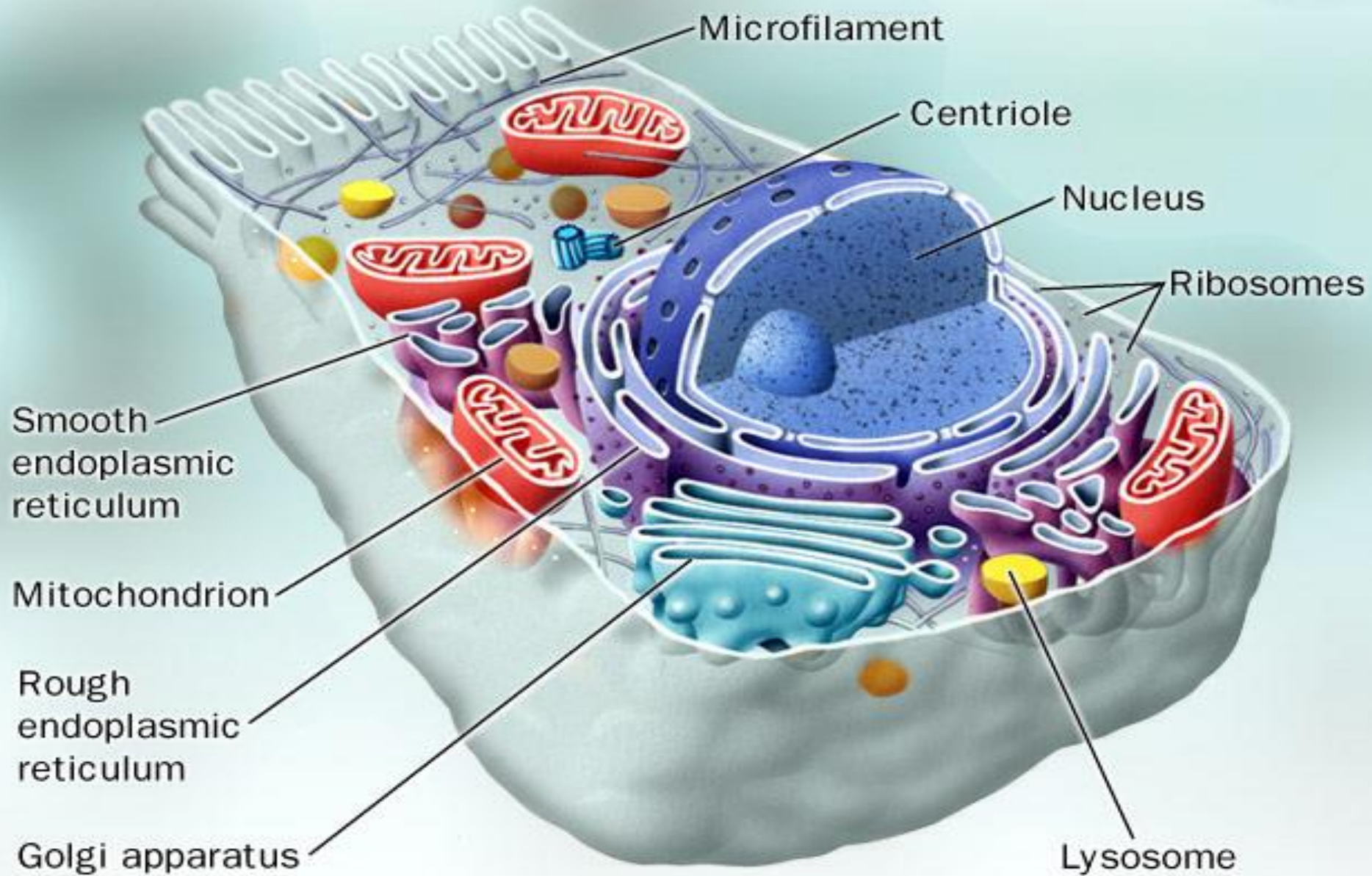
A plant cell

Animal cell as seen under the electron microscope.

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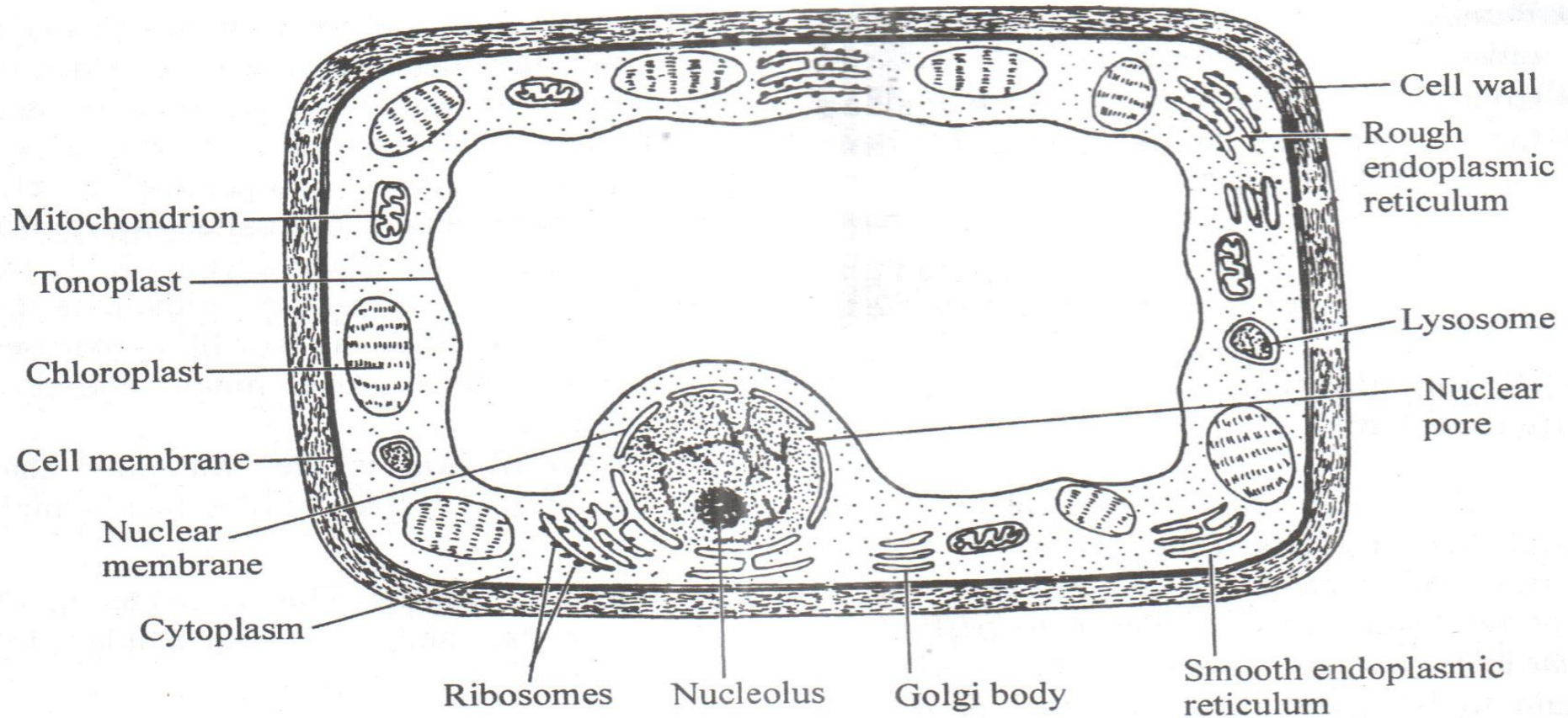


Animal cell as seen under the electron microscope



Plant cell as seen under electron microscope.

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Plant cell as seen under the electron microscope

Functions of cell organelles.

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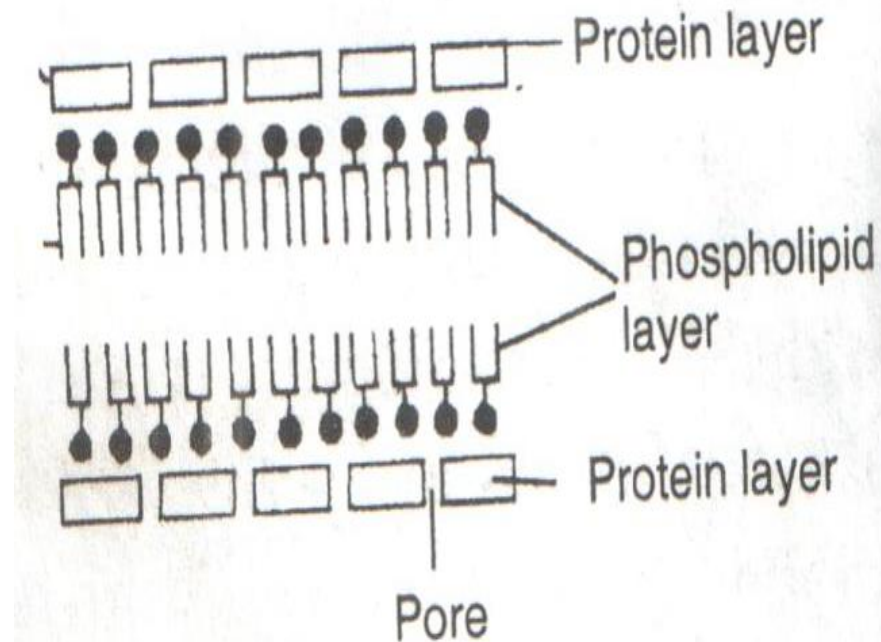
- **The cell** is the basic unit of life while **Cell organelles** are structures within the cells

1. THE CELL MEMBRANE/ PLASMA MEMBRANE/ PLASMALEMMA-

- It consists of three layers under the electron microscope. The three layers are composed of ***one layer of phospholipid*** found between ***two protein layers***.

Functions of the cell membrane.

- i. Enclose the cell contents.
- ii. Controls the movement of materials in and out of the cell hence said to be ***semi-permeable/selectively permeable***.



2. Cytoplasm

Functions/ roles.

- i. It is fluid medium in which chemical reactions take place.
 - ii. It contains other cell organelles and dissolved substances e.g. starch, glycogen, fat droplets.
- The cytoplasm is not static but has movement called **cytoplasmic streaming.**
 - This movement is important because it distributes materials and cell organelles within the cell.

3. Mitochondria (Singular-mitochondrion)

Functions/ roles.

- i. They provide a site for respiration (to provide energy). Mitochondria are bound by two membranes.

Structure

- The inner membrane is greatly folded into ***cristae*** to increase the surface area for attachment of respiratory enzymes.
- Has the **matrix** to provide a site for respiration.

- N/B Actively/ rapidly respiring cells e.g. muscle cell, sperm cell, apical meristems, kidney cell have numerous mitochondria. This is because they require a lot of energy.

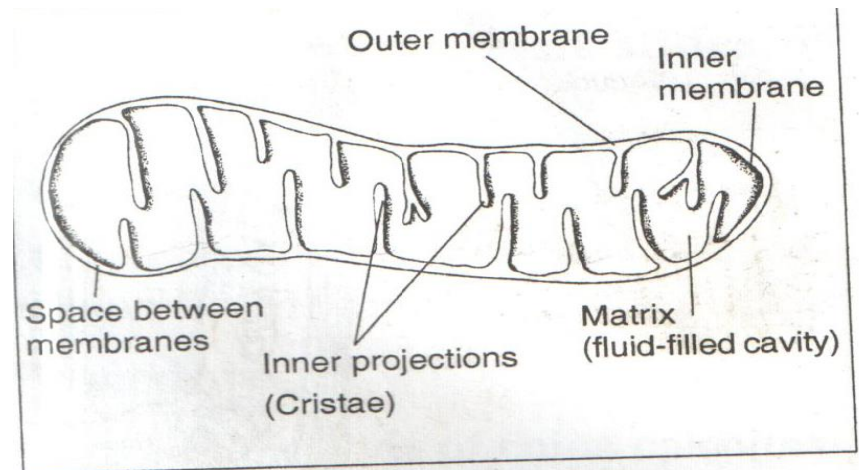


Fig. 3.6: The mitochondrion

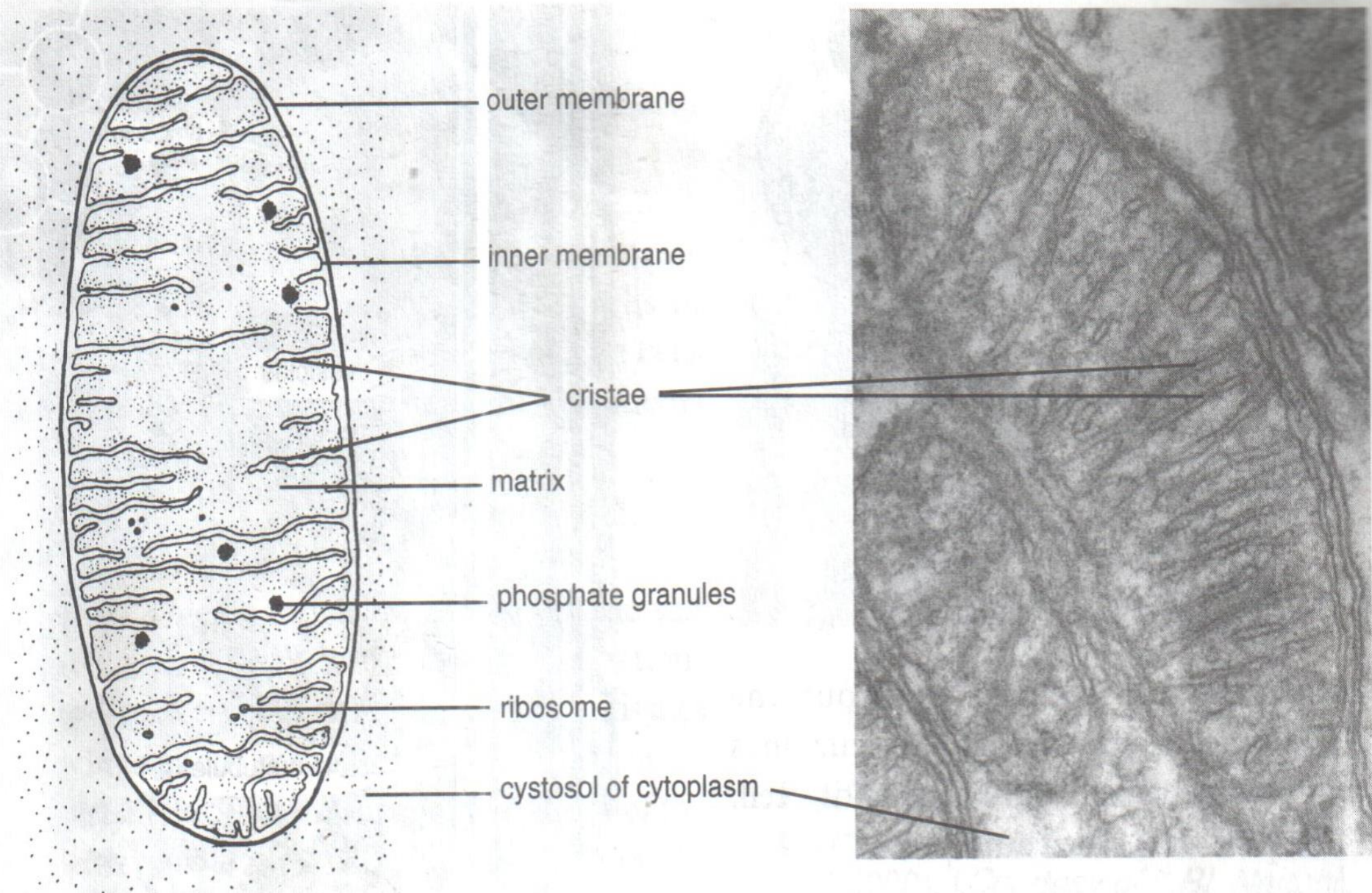


Fig. 2.9: L.S. showing internal structure of a mitochondrion x80,000 (micrograph courtesy of ILRI Nairobi).

4. **Endoplasmic Reticulum (ER)**- they are **interconnected channels** continuous with the outer membrane of the **nuclear membrane**.

- They include:
 - i. Smooth endoplasmic reticulum- they lack ribosomes on the surface.
 - ii. Rough endoplasmic reticulum- have ribosomes on the surface.

Functions.

- i. The rough endoplasmic reticulum **transports proteins**.
- ii. The smooth endoplasmic reticulum **manufactures/ synthesizes and transports lipids**.

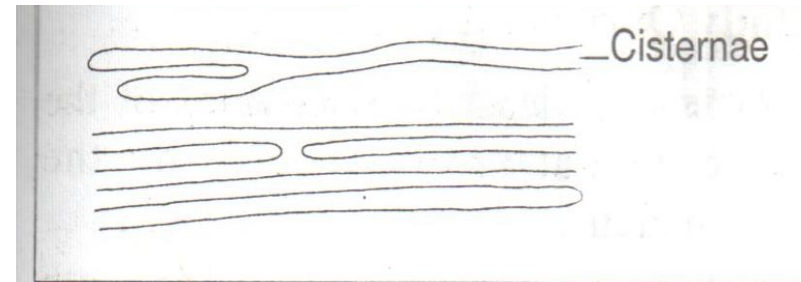


Fig. 3.7 (a): Smooth Endoplasmic reticulum

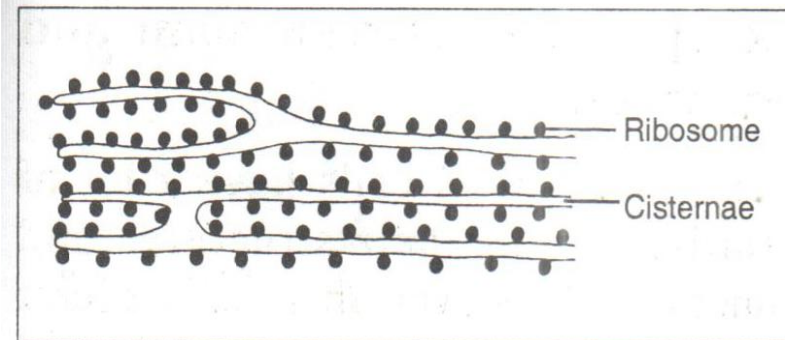


Fig. 3.7 (b): Rough Endoplasmic reticulum

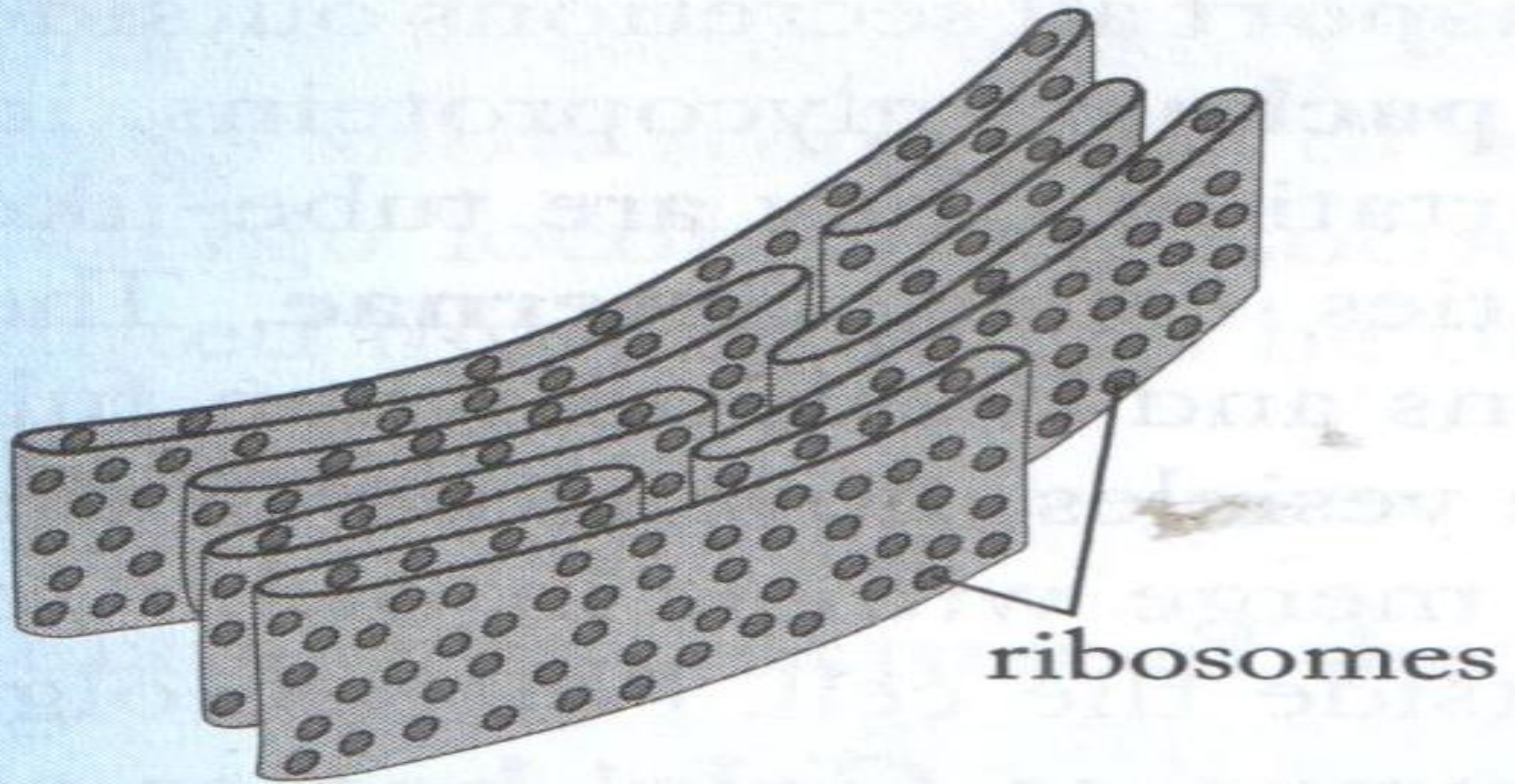


Fig 1.5: Rough Endoplasmic reticulum (rER)

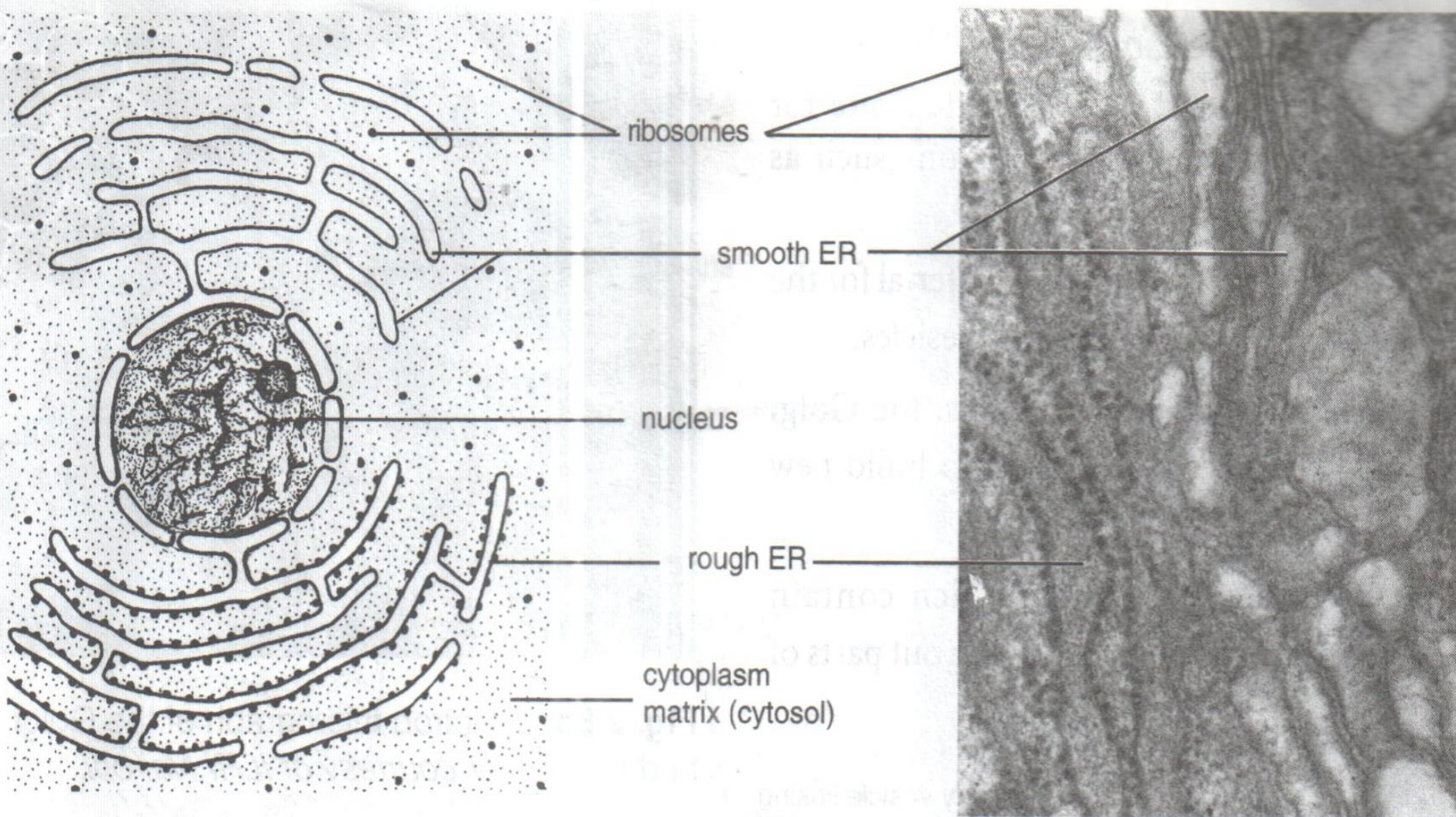


Fig. 2.7a: Rough (RER) and smooth endoplasmic reticulum (SER).

b: Electron micrograph showing RER and SER of part of a cell (x60,000).

5. **Ribosomes-** Ribosomes are spherical in shape.
- Some are scattered within the cytoplasm while others are bound to the surface of rough endoplasmic reticulum.

Function.

- i. They form sites for **protein synthesis**.

6. **Lysosomes.**

Functions

- i. Contain lytic enzymes which destroy worn out organelles/cells/tissue.
- ii. Digestion of food/bacteria.
- White blood cells contain numerous lysosomes because they destroy pathogens hence contain numerous lysosomes because they contain lytic enzymes that destroy pathogens.

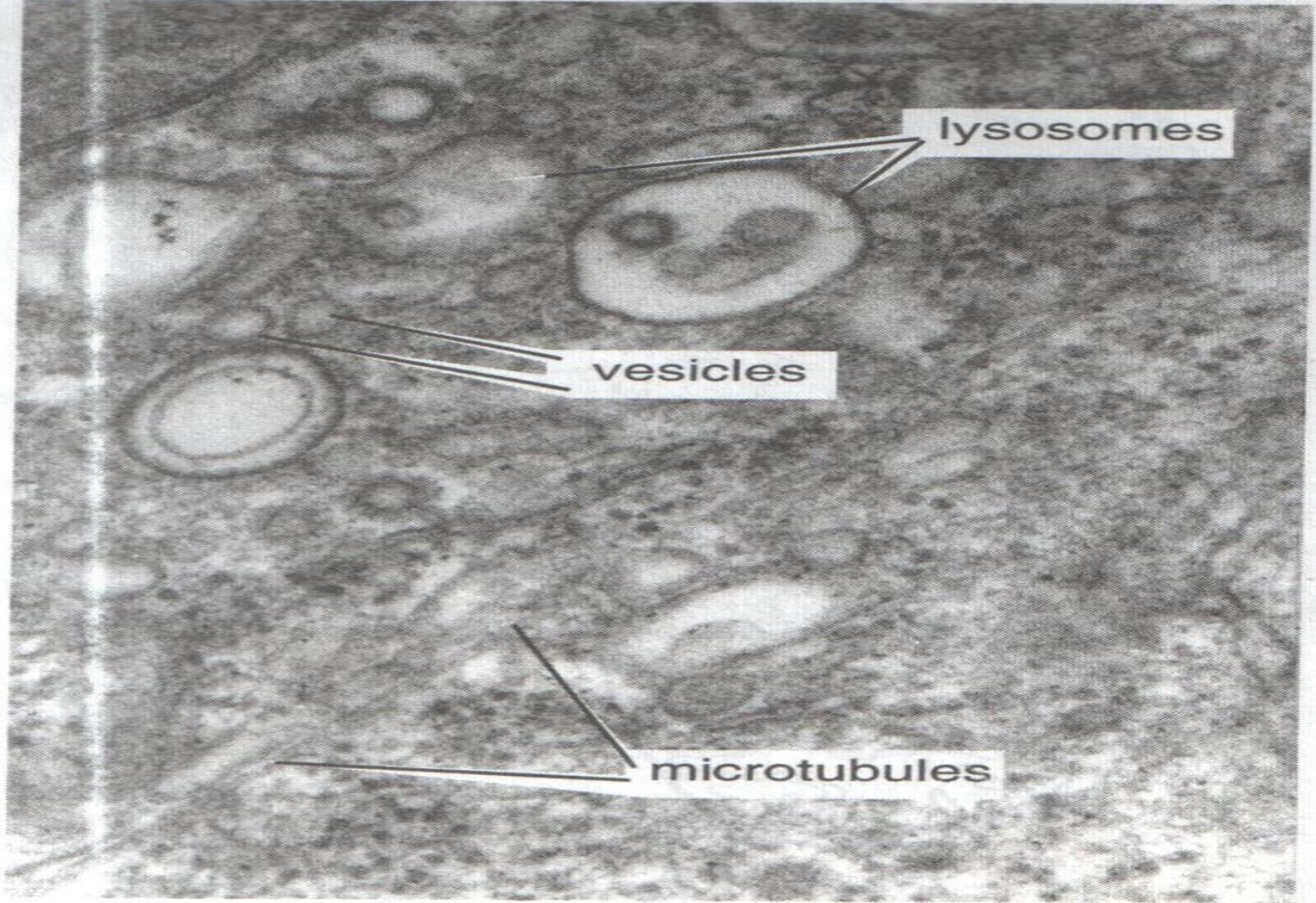


Fig. 2.10b: Electron micrograph of lysosomes.

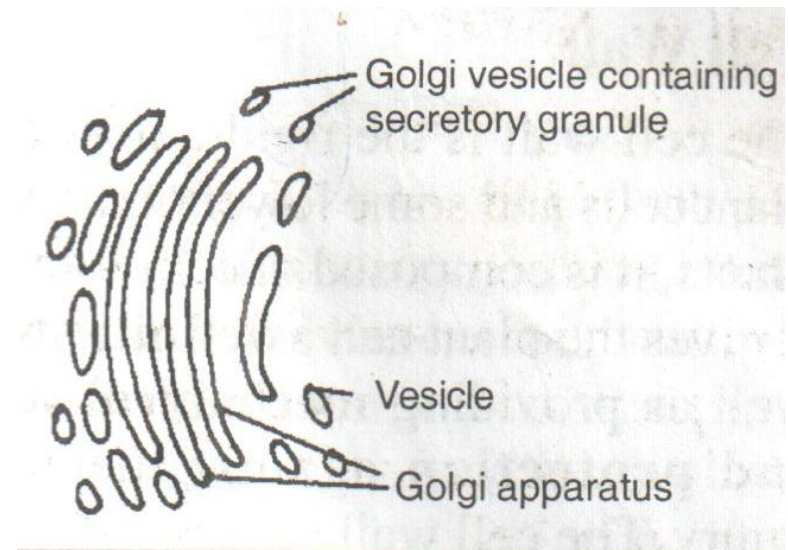
7. **Golgi bodies (Golgi apparatus)**- they are stacks of membrane-bound **tube like sacs**.

□ They are found close to the cell membrane.

Functions of golgi apparatus/bodies

- i. Packaging and transport of glycoproteins (glycogen/ carbohydrates and proteins) as secretions.
- ii. Transport of synthesized materials/substances out of the cell as secretions e.g. enzymes.

- iii. Secretion of synthesized/ manufactured glycoproteins (proteins and carbohydrates).
- iv. Formation of lysosomes.



The golgi apparatus

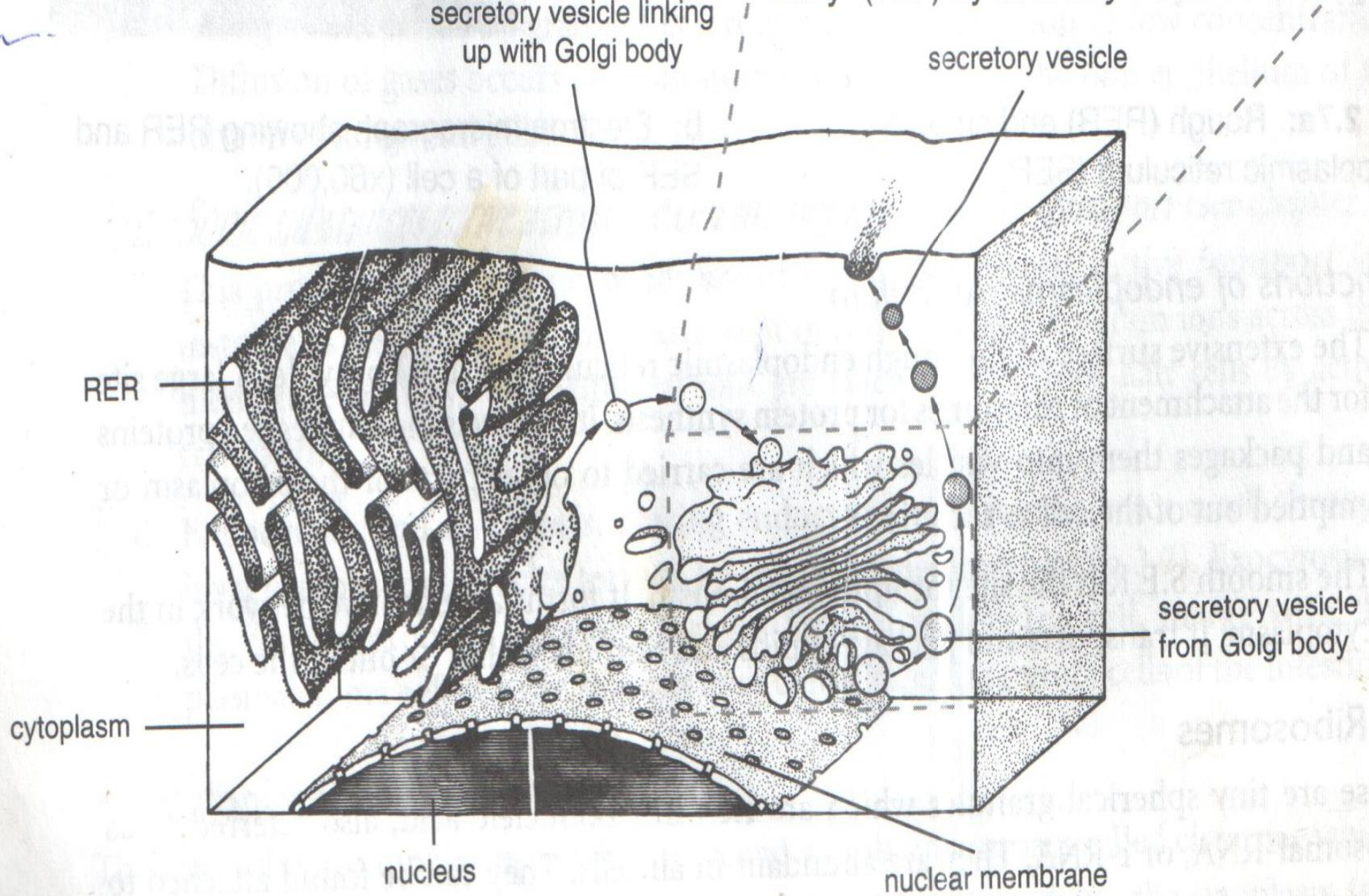


Fig. 2.8a: Three-dimension drawing showing a Golgi body in action.

8. **Centrioles-** they are rod-shaped structures located just outside the nuclear membrane.

Functions.

- i. They form spindle fibres during **cell division** in animal cells.

- ii. They help in the **formation** of **cilia** and **flagella** in cells and organisms where those structures occur e.g. protocista.

9. **Chloroplasts**- they are found in plant cells only.

Function.

- i. They provide a site for photosynthesis/ they contain chlorophyll that traps light for photosynthesis.

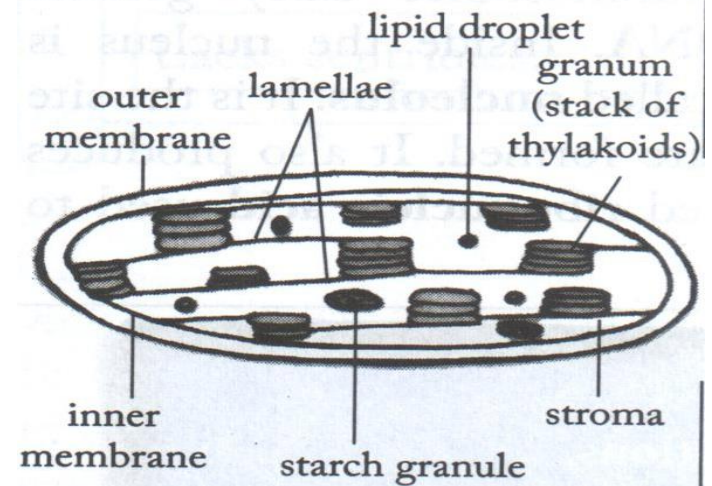


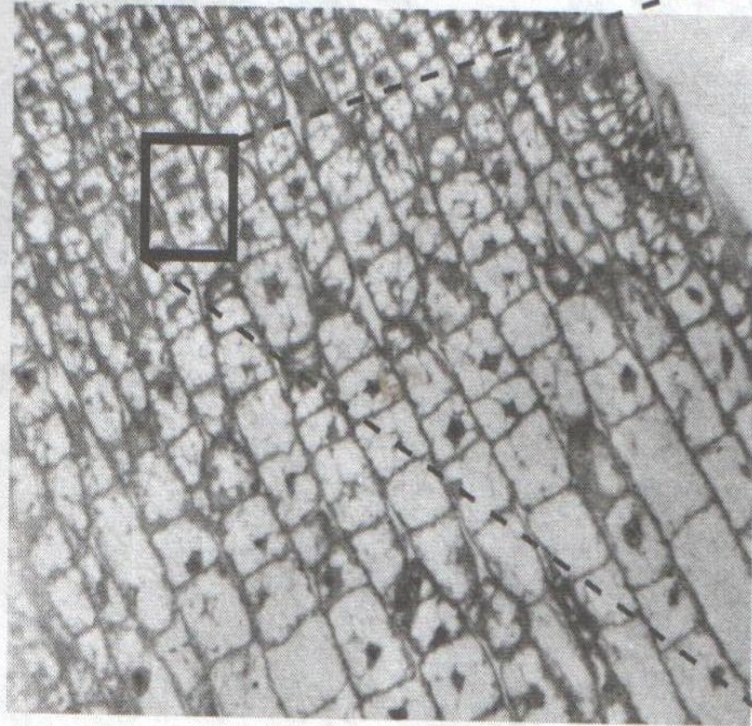
Fig 1.4: Chloroplast

10. **Vacuoles-** they are sacs which are filled with a fluid and vary in size.
- Animal cells contain small vacuoles which are small and temporary while plant cells contain large and permanent vacuoles.
 - In plants vacuoles are centrally placed and surrounded by a membrane called **tonoplast**.
 - They contain a solution of salt and sugars called cell sap hence they are called sap vacuoles.

Functions/ roles.

- i. They store salts and sugars.
- ii. The sap contributes to the osmotic properties of the cell.
- In unicellular organisms:
 - i. Contractile vacuole removes/ excretes excess water (osmoregulation)
 - ii. Contractile vacuole removes/ excretes metabolic wastes.
 - iii. Food vacuole is used for feeding/ digestion/ storage of food.

a.



b.

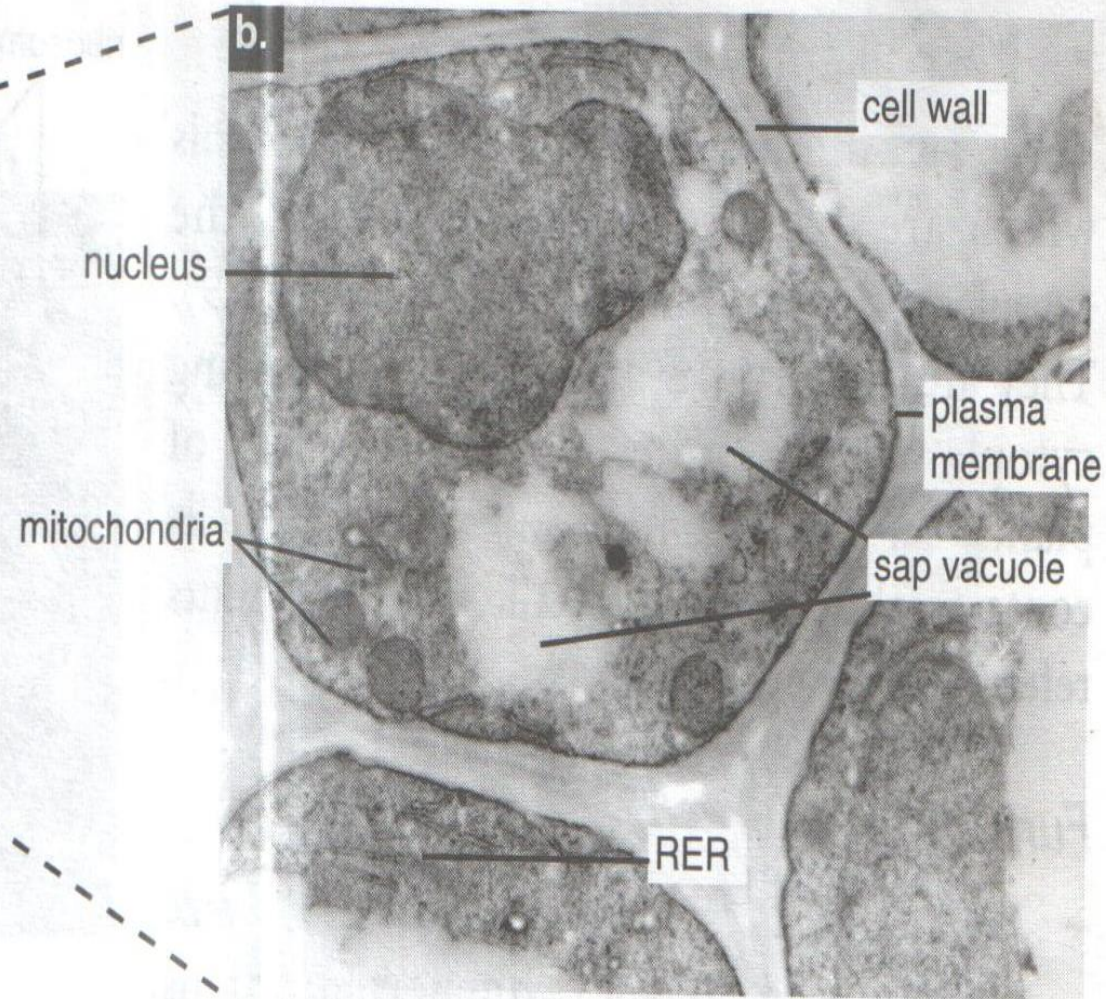


Fig. 2.13: a) Light micrograph showing plant cells (x500). **b)** Electron micrograph of plant cells in the root tip (x2,200) (*Courtesy of ICIPE, Nairobi*).

11. **Cell wall-** is is a membrane found in cells of plants and fungi; and in some protocists but **not** in animal cells.
- It is made up of **cellulose** in plant cells and either cellulose or **chitin** in fungi. It is rigid and tough.

Functions.

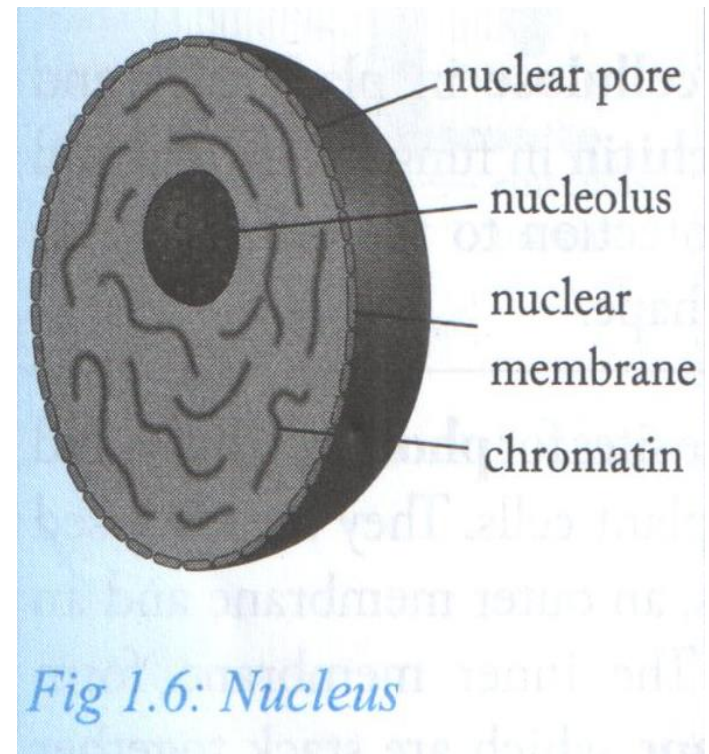
- i. Protects the cell against mechanical injury.
- ii. It gives the cell a definite shape.
- iii. Provides mechanical support

12. **Nucleus (plural-nuclei)-** contains:

- i. **Nuclear membrane** which has **Nuclear pore/ nucleopore-** which allows and controls the movement of substances in and out of the nucleus.
- ii. **Nucleoplasm** – it is a viscous fluid which has nucleolus and chromatin.
- iii. **Nucleolus-** manufactures the ribosomes.
- iv. **Chromatin-** contains hereditary/ genetic materials/ DNA.

Function of nucleus

- i. The nucleus controls all the cell activities e.g. cell division, protein synthesis, respiration, cell secretion, excretion and cell growth.
- **N/B A cell without a nucleus would only survive for a short time because it is not able to carry out normal cell functions.**



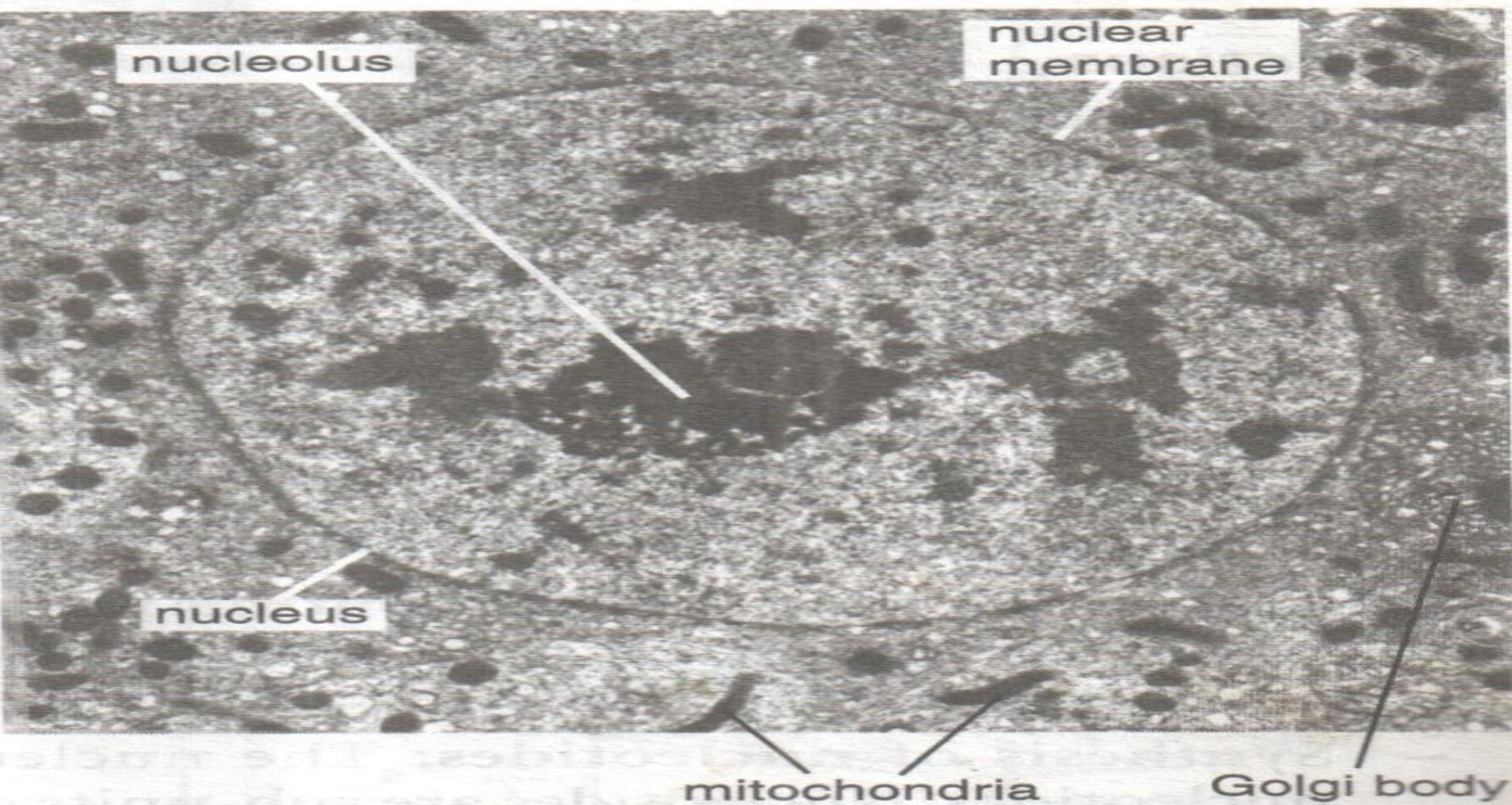


Fig. 2.12: EM of part of animal cell showing a nucleus with its nucleolus (x16,000)
(*Courtesy of ICIPE, Nairobi*)

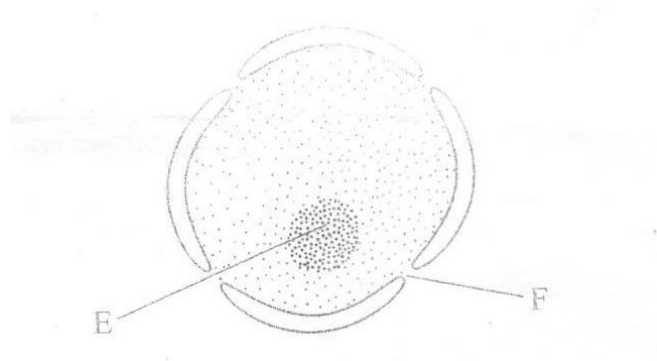
DIFFERENCES BETWEEN PLANT AND ANIMAL CELLS.

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Plant cell.	Animal cell
1. It is usually large.	1. It is usually smaller.
2. It is regular in shape.	2. It is irregular in shape.
3. It has a cell wall.	3. It lacks a cell wall.
4. It has a large central vacuole.	4. It has no vacuoles but if they are present , they are small and temporary and scattered within the cytoplasm.
5. The cytoplasm and nucleus are located towards the periphery.	5. The cytoplasm occupies most space in the cell with the nucleus centrally placed.
6. It has chloroplasts.	6. It lacks chloroplasts.
7. It stores starch, oils and proteins.	7. It stores glycogen and fats.
8. It lacks centrioles.	8. It has centrioles.

STUDY QUESTION

1. The diagram below represents a nucleus.



- a) Name the structures labeled E and F.

E- Nucleolus.

F- Nuclear pore/ Nucleopore.

- b) State the function of F.

☐ ***Allows/ facilitates the movement of materials in and out of the nucleus.***

2. State two main functions of the vacuole.

i. Osmoregulation/ removal of excess water (by contractile vacuole)

ii. Feeding/ digestion of food (by food vacuole)

iii. Excretion/ removal of metabolic wastes.

3. Name the plant cell organelle:

i) That stores chlorophyll- ***Chloroplast***

ii) Responsible for intracellular digestion- ***Lysosome.***

PREPARATION OF TEMPORARY SLIDES.

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- ☐ To observe specimen under the microscope, we need to prepare slides.
- ☐ There are two types of slides, namely:
 - a) **Temporary or fresh slides-** for immediate use during a laboratory exercise and
 - b) **Permanent slides-** which can be preserved for reuse.

Procedures carried out when preparing temporary slides.

1. **Sectioning-** It refers to cutting/ making of thin sections **to allow light to pass through or make them transparent.**
- ☐ It is done by use of a **sharp razor blade** to **ensure that cells are not distorted.**

2. **Adding a drop of water/ placing the sections in water-** to keep the cells turgid and prevent dehydration.
3. **Staining-** to make different parts of the cell more distinct and clear. Stains commonly used are **iodine solution, methylene/bromothymol blue, neutral red and eosin.**
4. **Mounting-** This is putting the specimen on the slide in the appropriate medium before covering it with a cover slip.

Advantages of using/placing a coverslip over the specimen.

1. To hold the specimen in place.
2. Protects specimen from dehydration/drying up/dust particles.
3. Protect the objective lens.
 - The stage should be kept dry for easy manipulation of specimen as wetness causes the specimen to stick onto the stage.

5. **Fixation**- It helps to make the specimen hard or stiff for sectioning.
 - It can also be done after sectioning to help maintain the structure of the specimen.
 - Commonly used fixative is **70% *ethanol***. Plant materials are stiff enough and do not require fixation.

Practical Activity

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Aim: To prepare and observe temporary slide of onion cells under the light microscope.

Requirements:

1. Microscope
2. Clean microscope slides
3. Cover slips
4. Scalpel or a new razor blade
5. Distilled water
6. Iodine solution.
7. Onion bulb.
8. Pair of forceps.
9. Dropper.
10. Mounted needle.

Procedure

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1. Cut the onion bulb vertically into four parts.
2. Separate a fleshy leaf from one of the parts.
3. Remove/ peel a thin piece of epidermis from this leaf using forceps. Trim down the epidermis to 5mm long.
4. Place a drop of iodine at the centre of a clean slide. Add a drop of water to dilute it.
5. Quickly spread the piece of epidermis onto the drop of water.
6. Using a mounted needle, lower a clear cover slip on to the epidermis strip. Do this gently to avoid trapping air bubbles.
7. Examine this temporary slide under the low and medium power objective lenses of microscope.

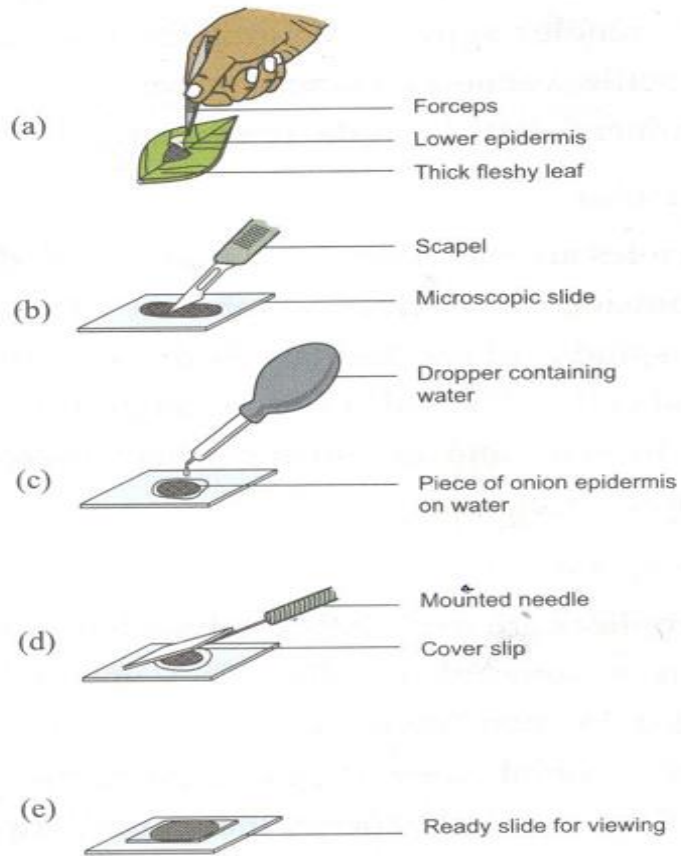


Fig 3.9 (a – e): Preparation of temporary slide of onion epidermis.

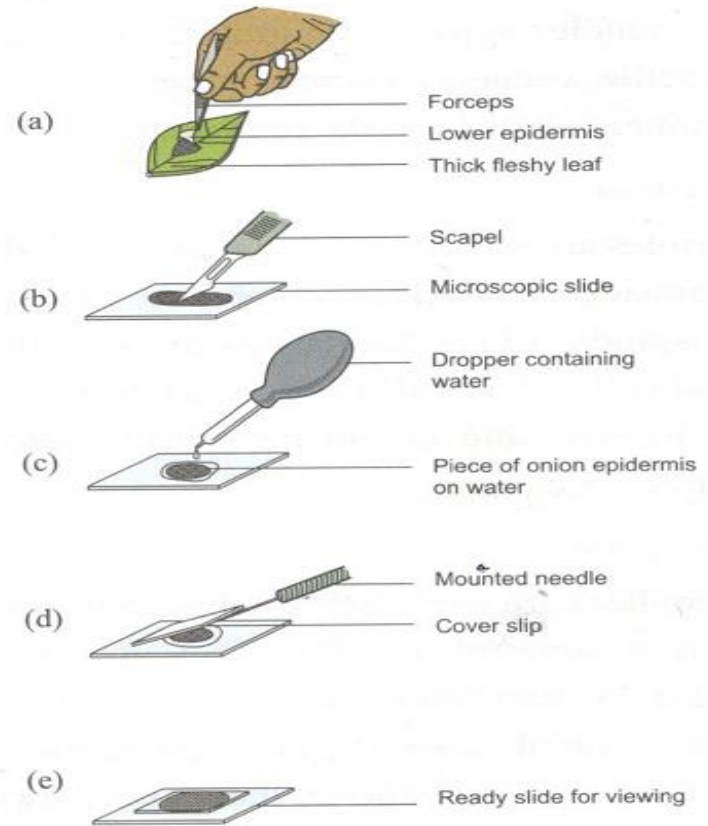


Fig 3.9 (a – e): Preparation of temporary slide of onion epidermis.

Estimation of cell size.

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- Most cells are shorter than a millimetre and therefore, their sizes are measured in smaller units called micrometres (μm).
- 1 millimetre (mm) = 1 000 micrometres/microns (μm).
- 1 (μm) = 1 000 nanometres (nm)

Practical activity

Aim: To estimate the size of onion epidermal cells.

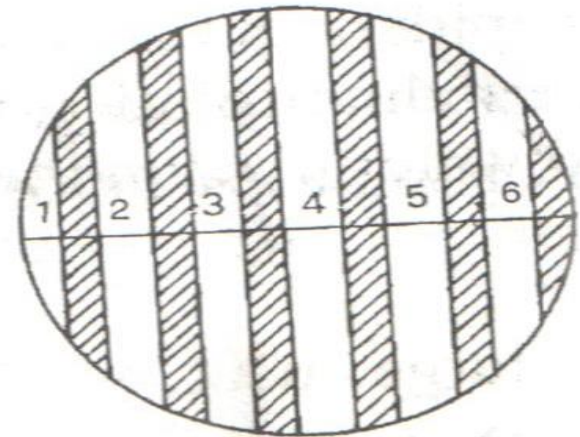
Requirements:

- i. Microscope.
- ii. Transparent ruler marked in millimetres.
- iii. Prepared slide of onion epidermis.

Procedure.

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1. With the low power objective lens in place, keep a transparent ruler on the stage of the microscope.
2. Focus so that the millimetres marks on the ruler are seen as thick dark lines.
3. Estimate the diameter of the field of view by counting the one-millimeter spaces between the first mark and the last one across the field of view e.g. from the figure below the diameter of field of view is 6.0 mm.



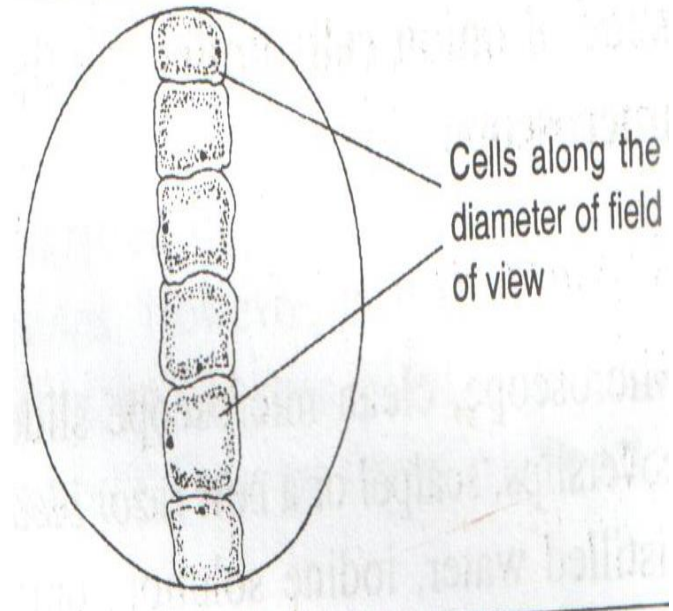
Diameter of field of view

4. Convert the diameter of the field of view from millimetres to micrometres.

□ $1 \text{ mm} = 1\,000 \mu\text{m}$

$$\begin{aligned} 6 \text{ mm} &= (6 \times 1\,000) \mu\text{m} \\ &= 6\,000 \mu\text{m} \end{aligned}$$

5. Remove the ruler and place the prepared slide of the onion epidermis.
6. Count the number of cells along the diameter of the field of view e.g. 6 cells as shown below.



7. Calculate the diameter of one cell using the following formula

$$\square \text{ Cell diameter} = \frac{\text{Diameter of field of view in } \mu m}{\text{No. of cells along the diameter the field of view.}}$$

Using the example above:

$$\text{Cell diameter} = \frac{6\,000}{6} = 1\,000 \mu m$$

6 cells

$$\square \text{ Therefore, diameter of 1 cell} = 1\,000 \mu m.$$

Limitations of using the light microscope to estimate the size of the cells.

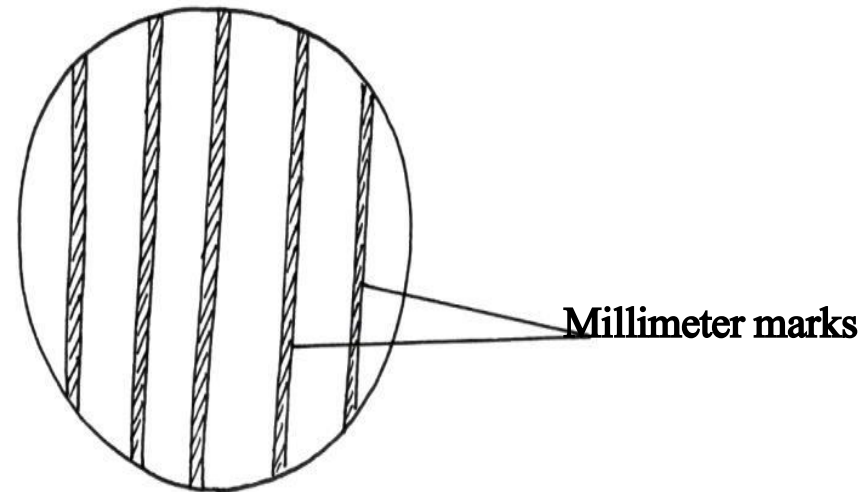
1. Cells vary in size and shape.
2. Cells in a tissue are not linearly arranged (not in a straight line)/cells are irregularly arranged.
3. Changes in osmotic pressure will affect the animal cells.

Other dimensions/parameters used to estimate the cell size.

- 1) Diameter.
- 2) Length.

Question 1.

- ☐ A student carried out an experiment on microscope work. The field of view was as shown in the following diagram.
- ☐ If she counted 20 cells on the diameter of the field of view, what was the approximate size of each cell in micrometers (μm). Show your working.



Solution

- Diameter of field of view

$$4 + \frac{1}{2} + \frac{1}{2} = 5 \text{ spaces}$$

$$\text{space} = 1 \text{ mm}$$

$$1000 \mu\text{m} = 1 \text{ mm}$$

$$5000 \mu\text{m} = 5 \text{ mm}$$

- Cell size = Diameter of field of view

No. of cells

$$\text{Cell size} = \frac{5000 \mu\text{m}}{20 \text{ cells}}$$

20 cells.

$$= 250 \mu\text{m}.$$

Study question 2

- A student viewed and drew a plant cell of a diameter 4mm using a light microscope whose eyepiece lens was marked X1 and objective lens marked X5. How many cells were linearly arranged along the microscope's field of view whose diameter was 8mm. (show your work.)

Solution

1mm = 1 000 μm .

Total magnification= Cell drawing diameter

Actual cell diameter

$$X\ 5 = \underline{4\ 000}$$

X

$$x = \underline{4\ 000}$$

5

$$= 800\mu\text{m}$$

- Cell diameter= Field of view diameter(in micrometers)

No. of cells (Y)

$$800\ \mu\text{m} = \underline{8000}\ \mu\text{m}$$

Y

$$Y = \underline{8000}\ \mu\text{m}$$

$$800\ \mu\text{m}$$

$$= 10\ \text{cells};$$

Study question 3

- The diameter of the field of view was estimated to be 5mm under a certain magnification. 5 cells were observed along the diameter of the field of view. What was the diameter of one cell in microns (μm)?

Solution

$$1 \text{ mm} = 1000 \mu\text{m}$$

$$50 \text{ mm} = \frac{5 \times 1000}{1}$$

$$1$$

$$= 5,000 \mu\text{m}.$$

$$\text{Cell diameter} = \frac{\text{Diameter of field of view}}{\text{Number of cells.}}$$

$$\text{Number of cells.}$$

$$= \frac{5,000}{5}$$

$$5$$

$$= 1,000 \mu\text{m}.$$

Study Question 4.

A student used a microscope with x40 objective lens and x5 eye piece lens. He observed 5 Cells in the field of view which had 2.2mm radius.

- a) Calculate the area of field of view in square micrometers (μm^2).

$$1 \text{ mm} = 1\,000 \mu\text{m}.$$

$$2 \text{ mm} = 2 \times 1\,000 \mu\text{m}$$

$$= 2\,000 \mu\text{m}.$$

$$\text{Area} = \frac{22}{7} \times 2000 \times 2000$$

$$7$$

$$= 125714.29 \mu\text{m}^2$$

- b) What is the average size of the cell in micrometers?

$$\text{Size of the cell} = \frac{\text{diameter of field of view}}{\text{number of cells}}$$

$$= \frac{4\,000 \mu\text{m}}{5}$$

$$5$$

$$= 800 \mu\text{m}.$$

c) Estimate the actual size/length of the cell.

Total magnification= eye lens magnification x
objective lens magnification

$$= 5 \times 40$$

$$= \times 200$$

Total magnification= Size/Length of cell image

Actual length/size of cell (X)

$$200 = \frac{800 \mu\text{m}}{\times}$$

$$\times = \frac{800}{200}$$

$$= 4 \mu\text{m}.$$

Study question 5.

- Study the photomicrograph below. Calculate the actual size/diameter of the nucleus in microns (μm).

Solution

Measure the diameter of the organelle nucleus e.g. 35 mm

$$1 \text{ mm} = 1\,000 \mu\text{m}$$

$$35 \text{ mm} = 35\,000 \mu\text{m}.$$

Total magnification= Diameter of the nucleus photomicrograph.

Actual
length/diameter/size

$$2,200 = \underline{35\,000 \mu\text{m}}$$

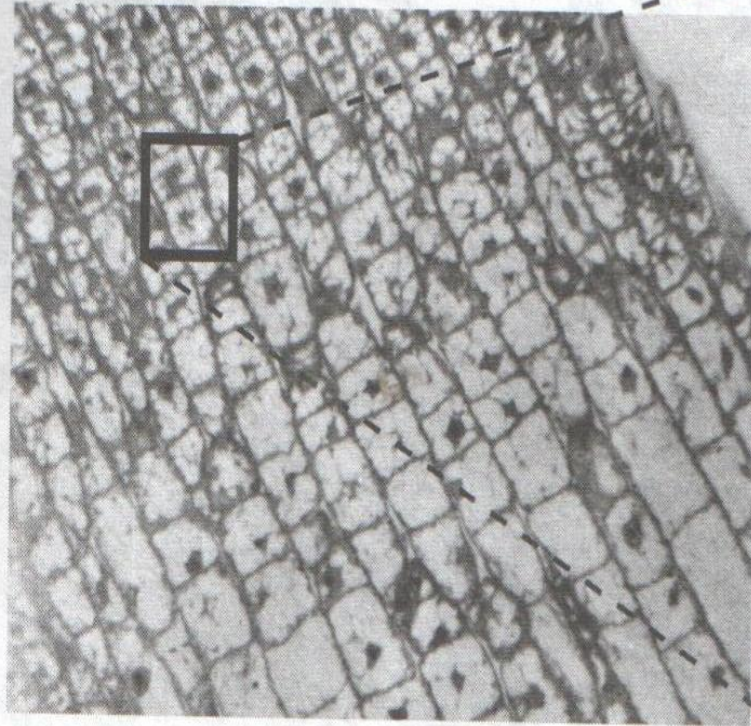
Actual size (Y)

$$Y = \underline{35\,000 \mu\text{m}}$$

$$2\,200$$

$$= 15.90 \mu\text{m}.$$

a.



b.

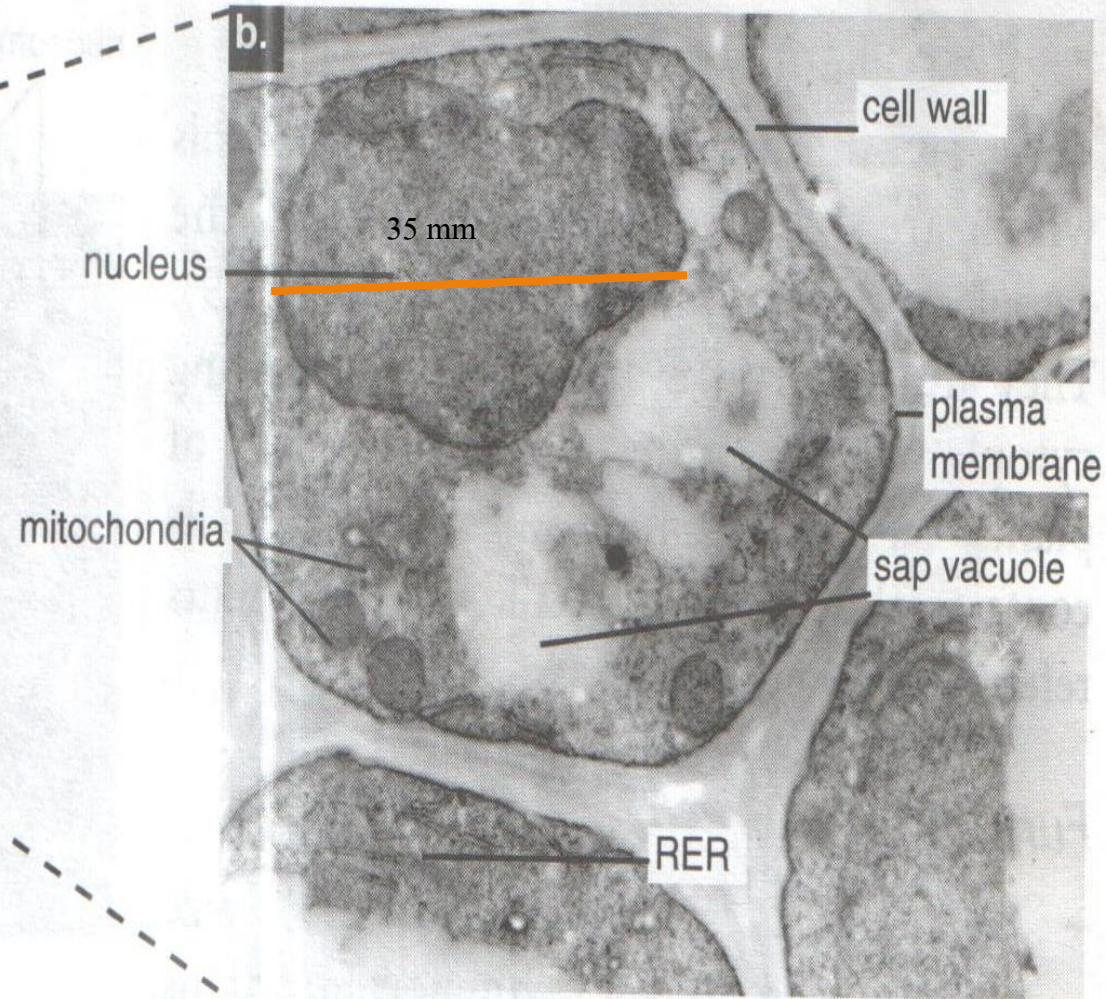


Fig. 2.13: a) Light micrograph showing plant cells (x500). **b)** Electron micrograph of plant cells in the root tip (x2,200) (*Courtesy of ICIPE, Nairobi*).

CELL SPECIALIZATION.

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- This is the structural modification of the cells to perform specific functions.

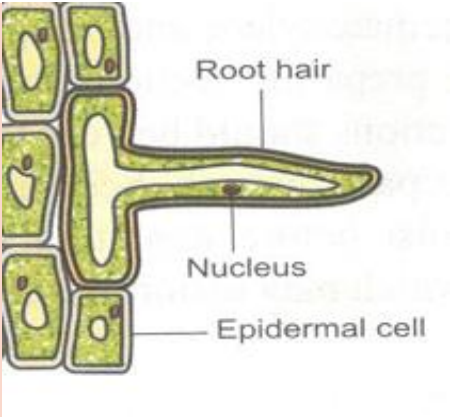
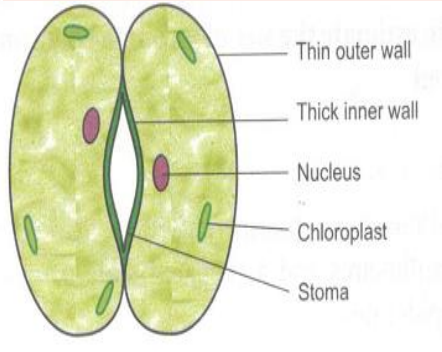
a) **Specialized plant cells**

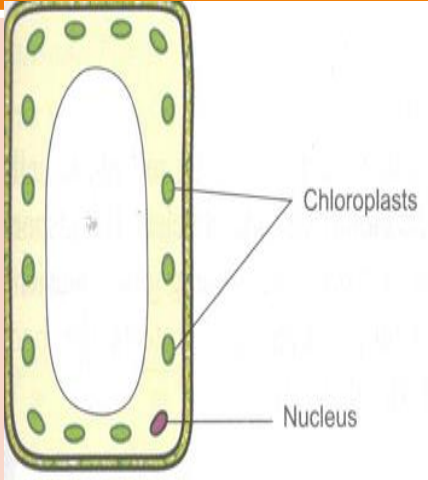
1. Root hair cell.
2. Guard cell.
3. Palisade cell.

b) **Specialized animal cell**

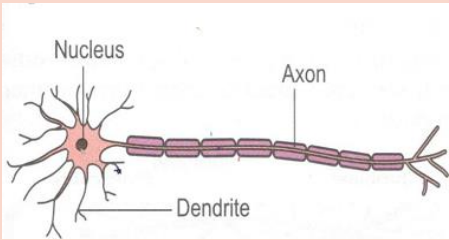
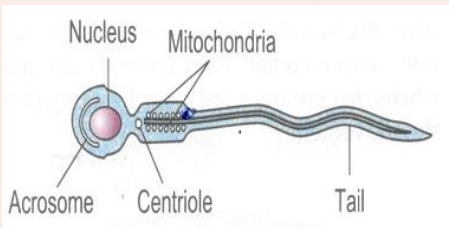
1. Nerve cell
2. Sperm cell.
3. Red blood cell.
4. White blood cell.
5. Muscle cell.



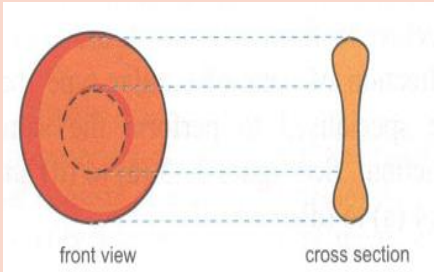
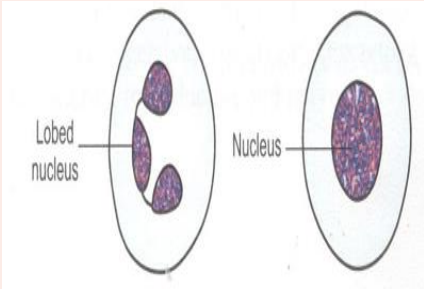
Specialized plant cell	Diagram	Structural modification/ adaptation	Function
1. Root hair cell		- It has extension/ root hair	-Increase surface area for absorption of water and mineral salts from the soil
2. Guard cell		- It has chloroplasts, thick inelastic inner wall and thin elastic outer wall	- Controls opening and closing of stomata. - Carries out photosynthesis.

Specialized plant cell	Diagram	Structural modification/ adaptation	Function
3. Palisade cell	 The diagram shows a rectangular palisade cell with a thick green cell wall. Inside, there are many small green oval chloroplasts arranged along the cell periphery. A large, clear central vacuole occupies most of the cell's interior. A single, dark purple nucleus is located near the bottom right corner. Labels with leader lines point to one of the chloroplasts and the nucleus.	- It contains numerous chloroplasts	- Photosynthesis.

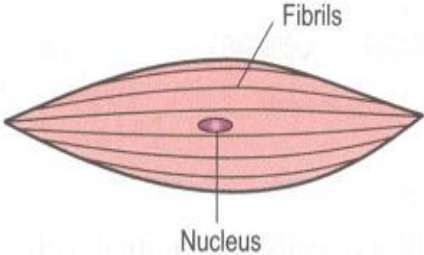
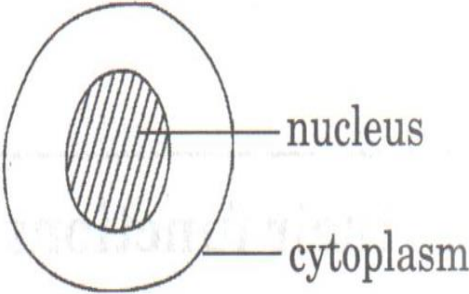


Specialized animal cell	Diagram	Structural modification/ adaptation	Function
1. Nerve cell		- It has axons and dendrites.	- Receive and transmit nerve impulses.
2. Sperm cell		- It has numerous mitochondria and long tail.	- Swim to reach and fertilize egg cell/ ovum



Specialized animal cell	Diagram	Structural modification/ adaptation	Function
3. Red blood cell	 The diagram illustrates the structure of a red blood cell. On the left, a 'front view' shows a red, biconcave disc with a dashed circle in the center. On the right, a 'cross section' shows the cell's thickness, highlighting its biconcave shape. Labels 'front view' and 'cross section' are at the bottom.	- It has biconcave shape, contain haemoglobin and lacks nucleus.	- Transports oxygen and carbon (IV) oxide within the body.
4. White blood cell	 The diagram shows two views of a white blood cell. On the left, a 'Lobed nucleus' is shown as three purple, kidney-shaped lobes within a cell. On the right, a 'Nucleus' is shown as a single, large, purple oval within a cell. Labels 'Lobed nucleus' and 'Nucleus' are on the left.	- It has large nucleus, show amoebic movement/ are able to change shape.	- Protect the body against infections.



Specialized animal cell	Diagram	Structural modification/ adaptation	Function
5. Muscle cell.		- It has contractile fibrils.	- Brings about movement.
6. Egg cell/ ovum		- Large cytoplasm.	- Stores food for developing embryo.

TISSUES

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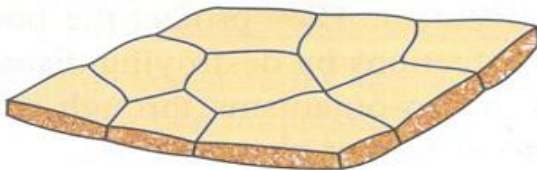
A tissue is a group/ collection of cells that are specialized to perform similar functions

Animal tissues.

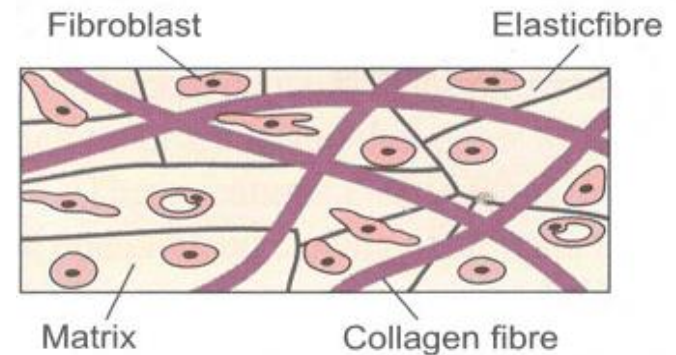
1. **Epithelial tissue-** it consists of epithelial cells that form layers. They are found on the outside of the body or around internal organs.

Function.

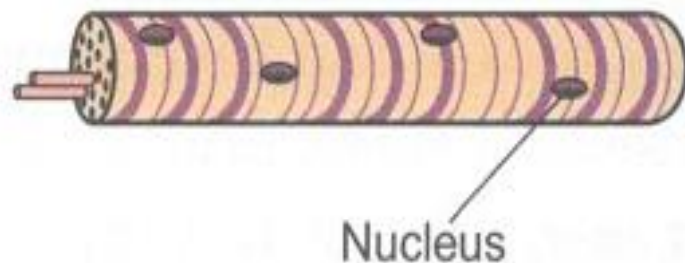
- i. It protect the internal and external surfaces.



2. **Connective tissue-** consists of strong fibres that connect other tissues and organs holding them together in position.



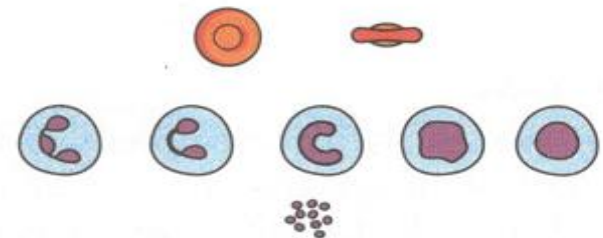
3. **Skeletal tissue-** it consists of elongated cells with fibres that contracts and relax to bring about movement.



4. **Blood tissue-** it consists of platelets, white blood cells and red blood cells.

Function.

- i. Protects the body against infection/diseases.
- ii. Transport materials in the body e.g. oxygen, metabolic wastes and nutrients.

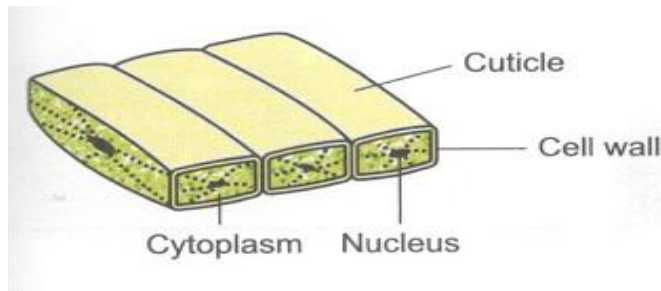


PLANT TISSUES

1. **Epidermal tissue-** It consists of a single layer of epidermal cells covering the outer surface of leaves and on young parts of the stem and roots.

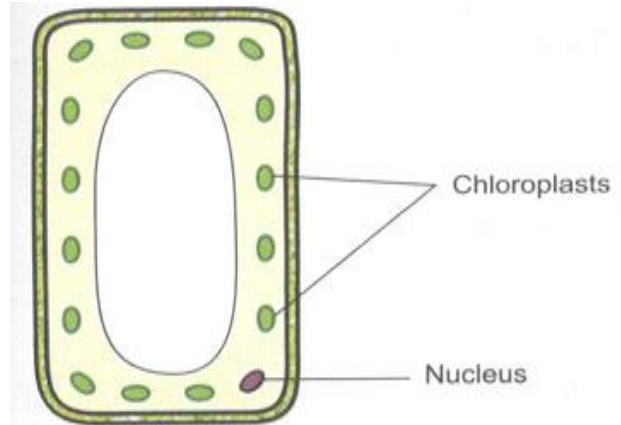
Function

- i) It protects inner tissues of plant from mechanical damage and infection.

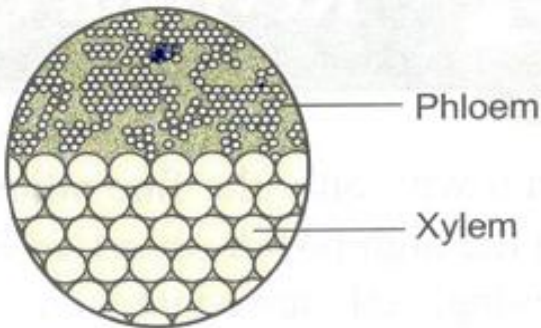


2. **Palisade / photosynthetic tissue.**

- ☐ It consists of numerous chloroplasts to trap light energy for photosynthesis.



3. **Vascular tissue-** It is composed of **xylem** which transports water from the roots to the leaves and **phloem** which transports manufactured food from the leaves to other parts of the plant.



4. **Meristematic tissue-** It consists of meristematic cells found at growing regions of plants that **actively divide to allow growth.**
- ☐ Apical meristems bring about increase in height of roots and shoots.
 - ☐ Lateral meristem bring about secondary thickening.

5. Parenchyma tissue.

- It consists of thin walled and irregularly shaped cells.

Role/ function.

- Provide mechanical support (packaging) and storage of substances.

ORGANS

- An organ is a group of tissues that are specialized to perform similar/same function.
- Examples of animal organs include: **stomach, brain, kidney, liver, heart, eye, ear e.t.c.**
- Examples of plant organs include: **roots, leaves, flowers, stem e.t.c.**

ORGAN SYSTEM

- An organ system is a group of organs that perform the same/similar function.
- Examples of animal organ systems include: digestive system, circulatory system, respiratory system, excretory system, nervous system, reproductive system e.t.c.

□ Example in plants is transport system.

